

Hints & Solutions

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Q1 (1) - Domain and Range

Given, $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$

To find domain, we must have

$$\sin^{-1}(2x) + \frac{\pi}{6} \geq 0 \quad \left[\text{but } \frac{-\pi}{2} \leq \sin^{-1} \theta \leq \frac{\pi}{2} \right]$$

Now, $-\frac{\pi}{6} \leq \sin^{-1}(2x) \leq \frac{\pi}{2}$

$$\Rightarrow \sin\left(\frac{-\pi}{6}\right) \leq 2x \leq \sin\left(\frac{\pi}{2}\right)$$

$$\Rightarrow -\frac{1}{2} \leq 2x \leq 1$$

$$\Rightarrow -\frac{1}{4} \leq x \leq \frac{1}{2}$$

$$\therefore x \in \left[-\frac{1}{4}, \frac{1}{2}\right]$$

Q2 (2) - Domain and Range

We have, $f(\cos 4\theta) = \frac{2 \cos^2 \theta}{2 \cos^2 \theta - 1} = \frac{1 + \cos 2\theta}{\cos 2\theta} \dots (i)$

when, $\cos 4\theta = \frac{1}{3}$

$$\Rightarrow 2 \cos^2 2\theta - 1 = \frac{1}{3} \Rightarrow 2 \cos^2 2\theta = \frac{4}{3}$$

$$\Rightarrow \cos^2 2\theta = \frac{2}{3}$$

$$\therefore \cos 2\theta = \pm \sqrt{\frac{2}{3}}$$

Thus, $f\left(\frac{1}{3}\right) = \frac{1 \pm \sqrt{\frac{2}{3}}}{\pm \sqrt{\frac{2}{3}}} = \frac{\sqrt{3} + \sqrt{2}}{\pm \sqrt{2}}$

$$\therefore f\left(\frac{1}{3}\right) = \sqrt{\frac{3}{2}} + 1 \text{ or } -\sqrt{\frac{3}{2}} + 1$$

Q3 (2) - Domain and Range

Let $y = \frac{x^2 + 14x + 9}{x^2 + 2x + 3}$

$$\Rightarrow x^2 y + 2xy + 3y = x^2 + 14x + 9$$

$$\Rightarrow x^2(y - 1) + x(2y - 14) + 3y - 9 = 0$$

$$\therefore x \in R$$

$$\therefore \Delta \geq 0$$

$$\Rightarrow 4(y - 7)^2 - 4 \cdot (y - 1)3(y - 3) \geq 0$$

$$\Rightarrow y^2 + 49 - 14y - 3(y^2 - 4y + 3) \geq 0$$

$$\Rightarrow -2y^2 - 2y + 40 \geq 0$$

$$\Rightarrow y^2 + y - 20 \leq 0$$

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$$\Rightarrow (y+5)(y-4) \leq 0$$

$$\Rightarrow -5 \leq y \leq 4$$

$$\therefore \text{Range} = [-5, 4]$$

Q4 (2) - Domain and Range

$$\text{We have, } f(x) = \log_{\sqrt{5}} \{ \sqrt{2}(\sin x - \cos x) + 3 \}$$

$$\text{Since, } -\sqrt{2} \leq \sin x - \cos x < \sqrt{2}$$

$$\Rightarrow 1 \leq \sqrt{2}(\sin x - \cos x) + 3 \leq 5$$

$$\Rightarrow \log_{\sqrt{5}} 1 \leq \log_{\sqrt{5}} \{ \sqrt{2}(\sin x - \cos x) + 3 \} \leq \log_{\sqrt{5}} 5$$

$$\Rightarrow 0 \leq f(x) \leq 2$$

$$\therefore \text{Range} = [0, 2]$$

Q5 (2) - Identical Functions

For identical functions domain and range should be same.

$$(1) f(x) = \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \text{ for all } -1 \leq x \leq 1$$

$$g(x) = \pi/2 \text{ for all } x \in R$$

$\therefore f$ and g are not identical.

$$(2) f(x) = \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2} \text{ for all } x \in R$$

$$g(x) = \frac{\pi}{2} \text{ for all } x \in R$$

$\therefore f$ and g are identical.

$$(3) f(x) = \sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2} \text{ for all } |x| \geq 1.$$

$$g(x) = \frac{\pi}{2} \text{ for all } x \in R$$

$\therefore f$ and g are not identical.

Q6 (2) - Identical Functions

$$(1) \text{ We have, } f(x) = \sqrt{x^2} = |x|, x \in R$$

$$\text{and } g(x) = (\sqrt{x})^2 = x, x \geq 0$$

Clearly, $f(x) \neq g(x)$ for all x , as

$$(2) \text{ We have, } f(x) = \sec(\sec^{-1} x) = x, |x| \geq 1$$

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and $g(x) = \operatorname{cosec}(\operatorname{cosec}^{-1} x) = x, |x| \geq 1$

Clearly, $f(x) = g(x)$ Hence, f and g are identical.

(3) We have, $f(x) = \sqrt{\frac{1+\cos 2x}{2}} = \sqrt{\frac{2\cos^2 x}{2}} = \sqrt{\cos^2 x} = |\cos x|$

and $g(x) = \cos x$

Clearly, $f(x) \neq g(x)$ for all x .

(4) We have, $f(x) = x$ and $g(x) = \log_e x = x, x > 0$ Then, $f(x) = g(x) = x$

Hence, f and g are not identical.

Q7 (4) - Identical Functions

(1) We have, $f(x) = \sqrt{1 + \sin x}$

$$= \sqrt{\left(\cos \frac{x}{2} + \sin \frac{x}{2}\right)^2} = \left|\cos \frac{x}{2} + \sin \frac{x}{2}\right|$$

and $g(x) = \sin \frac{x}{2} + \cos \frac{x}{2}$

Clearly, $f(x) \neq g(x)$ for all x

(2) We have, $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = 2 \tan^{-1} x$, if $-1 \leq x \leq 1$

and $g(x) = 2 \tan^{-1} x$

Thus, $f(x) \neq g(x)$

Hence, f and g are not identical in their common domain.

(3) We have, $f(x) = \sqrt{x^2} = |x|, x \in R$

and $g(x) = (\sqrt{x})^2 = x, x \geq 0$

Clearly, $f(x) \neq g(x)$ for all x

(4) We have,

$$\begin{aligned} f(x) &= \log_e x^3 + \log_e x^5 \\ &= \log_e (x^3 \cdot x^5) = \log_e x^8 = 8 \log_e x, x > 0 \end{aligned}$$

and $g(x) = 8 \log_e x, x > 0$

Hence, $f(x)$ and $g(x)$ are identical.

Q8 (2) - Periodic Function

We have, $f(x) = \sin^{-1}(\sin x)$

We know that, if $f(x)$ is not periodic and $g(x)$ is periodic, then $f \circ g(x)$ is periodic with period of $g(x)$ Hence,

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$f(x) = \sin^{-1}(\sin x)$ is periodic and its period $= 2\pi$

Q9 (1) - Periodic Function

We have, $f(x) = |\cos 2x|$

We know that, period of $|\cos x|$ is π .

$\therefore$ Period of $|\cos 2x| = \frac{\pi}{2}$

Q10 (1) - Periodic Function

We have, $f(x) = |\sin x| + |\cos x|$

Let $T_1 =$ Period of $|\sin x| = \pi$

and $T_2 =$ Period of $|\cos x| = \pi$

$\therefore f(x)$ is even function and both are complementary to each other.

$\therefore$ Its period is $\frac{1}{2}(\text{LCM of } T_1 \text{ and } T_2)$

$$= \frac{1}{2}\pi = \frac{\pi}{2}$$

Q11 (2) - Periodic Function

We have, $f(x) = \sin^4 x + \cos^4 x$

$$f(x) = (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x$$

$$= 1 - \frac{1}{2}(2\sin x \cdot \cos x)^2$$

$$= 1 - \frac{1}{2}\sin^2 2x$$

$$= 1 - \frac{1}{2}\left(\frac{1 - \cos 4x}{2}\right)$$

$$= 1 - \frac{1}{4}(1 - \cos 4x)$$

$$\therefore \text{ Its period } = \frac{2\pi}{4} = \frac{\pi}{2}$$

Q12 (3) - Periodic Function

We have, $f(x) = \sec(\sin x)$, $f(x + \pi) = \sec(\sin(x + \pi))$

$$= \sec(-\sin x)$$

$$= \sec(\sin x)$$

$$= f(x)$$

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Hence its period is π .

Q13 (4) - Periodic Function

We have, $f(x) = \sin(\sqrt{[a]}x)$

Its period = $\frac{2\pi}{\sqrt{[a]}} = \pi$ (given)

$$\Rightarrow \sqrt{[a]} = 2$$

$$\Rightarrow [a] = 4$$

$$\Rightarrow 4 \leq a < 5$$

$$\therefore a \in [4, 5)$$

Q14 (1) - Periodic Function

We have,

$$f(x) = (x + a) - [x + b] + \sin \pi x + \cos 2\pi x + \sin 3\pi x + \cos 4\pi x + \cdots + \sin(2n - 1)\pi x + \cos 2n\pi x$$

To period of $x + a - [x + b] + b - b$

$$\Rightarrow x + b - [x + b] + a - b$$

$$\Rightarrow \{x + b\} + a - b$$

Hence, we see that its period is 1.

Now, period of $\sin \pi x$ is $\frac{2\pi}{\pi} = 2$ Period of $\cos 2\pi x$ is $\frac{2\pi}{2\pi} = 1$

Similarly, period of $\cos 2n\pi x = \frac{2\pi}{2n\pi} = \frac{1}{n}$

$\therefore$ Period of $f(x)$ is LCM of all above period, which is 2.

Q15 (3) - Periodic Function

We have, $f(x) = \sin \frac{\pi}{4}[x] + \cos \frac{\pi x}{2} + \cos \frac{\pi}{3}[x]$

Let $T_1 =$ Period of $\sin \frac{\pi}{4}[x]$

$$\Rightarrow \sin \frac{\pi}{4}[x + T_1] = \sin \frac{\pi}{4}[x]$$

$$\Rightarrow \frac{\pi}{4}[x + T_1] = 2n\pi + \frac{\pi}{4}[x]$$

$$\Rightarrow [x + T_1] = 8n + [x]$$

$$\therefore T_1 = 8n$$

Its minimum value is 8.

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Hence, $T_1 = 8$

$$T_2 = \text{Period of } \cos \frac{\pi x}{2} = \frac{2\pi}{\frac{\pi}{2}} = 4$$

$$T_3 = \text{Period of } \cos \frac{\pi}{3}[x] = 6$$

$\therefore$ Period of $[x] = \text{LCM of } T_1, T_2 \text{ and } T_3 = 24$

Q16 (1) - Periodic Function

We have,

$$\begin{aligned} f(x) &= [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] - 3x + 15 \\ &= -x + [x] - \left(x + \frac{1}{3}\right) + \left[x + \frac{1}{3}\right] - \left(x + \frac{2}{3}\right) \\ &\quad + \left[x + \frac{2}{3}\right] + 15 + \frac{1}{3} + \frac{2}{3} \end{aligned}$$

$$\Rightarrow f(x) = -\{x\} - \left\{x + \frac{1}{3}\right\} - \left\{x + \frac{2}{3}\right\} + 16$$

$$\therefore f\left(x + \frac{1}{3}\right) = -\left\{x + \frac{1}{3}\right\} - \left\{x + \frac{2}{3}\right\} - \{x + 1\} + 16$$

$$= -\left\{x + \frac{1}{3}\right\} - \left\{x + \frac{2}{3}\right\} - \{x\} + 16 \text{ as } \{x + 1\} = \{x\}$$

$$\Rightarrow f\left(x + \frac{1}{3}\right) = f(x)$$

$\therefore$ Period of $f(x)$ is $\left(\frac{1}{3}\right)$.

Q17 (1) - Composite Functions

We have, $f : [-6, 6] \rightarrow R$

Then, $f(x) = x^2 - 3$ for $x \in R$

$$(\text{fofof})(-1) + (\text{fofof})(0) + (\text{fofof})(1) = ?$$

$$\therefore f \circ f(x) = f(x^2 - 3)$$

$$\begin{aligned} &= (x^2 - 3)^2 - 3 = x^4 + 9 - 6x^2 - 3 \\ &= x^4 + 6 - 6x^2 \end{aligned}$$

$$\text{and } (f \circ f \circ f)(x) = (x^4 + 6 - 6x^2)^2 - 3$$

$$\therefore (\text{fofof})(-1) = [(-1)^4 + 6 - 6(-1)^2]^2 - 3 = -2$$

$$(\text{fofof})(0) = 36 - 3 = 33$$

$$\text{and } (\text{fofof})(1) = (1 + 6 - 6)^2 - 3 = -2$$

$$\therefore (\text{fofof})(-1) + (\text{fofof})(0) + (\text{fofof})(1)$$

$$= -2 + 33 - 2 = 29$$

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Now, $f(y) = y^2 - 3 = 29$

$\Rightarrow y^2 = 32$

$\Rightarrow y = 4\sqrt{2}$

$\therefore (\text{fofof})(-1) + (\text{fofof})(0) + (\text{fofof})(1) = f(4\sqrt{2})$

Q18 (1) - Composite Functions

We have,

$$g(x) = x^2 + x - 2$$

and $\frac{1}{2}(g \circ f)(x) = 2x^2 - 5x + 2$

Now, $(g \circ f)(x) = 4x^2 - 10x + 4$
 $= (2x - 3)^2 - 5 + 2x$

and $g\{f(x)\} = (2x - 3)^2 + (2x - 3) - 2$

$\therefore f(x) = 2x - 3$

Q19 (4) - Composite Functions

We have,

$f(x) = \frac{x}{x+1}$ and $g(x) = \frac{x}{1-x}$

$\therefore (f \circ g)(x) = f\left(\frac{x}{1-x}\right) = \frac{\frac{x}{1-x}}{\frac{x}{1-x} + 1}$

$= \frac{\frac{x}{1-x}}{\frac{x+1-x}{1-x}} = x$

Q20 (2) - Composite Functions

Given,

$$g(x) = 1 + \{x\}, f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

where $\{x\}$ represents the fractional part function.

$\therefore f\{g(x)\} = \begin{cases} -1, & 1 + \{x\} < 0 \\ 0, & 1 + \{x\} = 0 \\ 1, & 1 + \{x\} > 0 \end{cases}$

$= 1, \forall x \in R, [\because 1 + \{x\} > 0]$

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Q21 (3) - Composite Functions

Given, domain of $f(x)$ is $[-5, 7]$

and $x \in [-5, 7]$

and $g(x) = |2x + 5|$

$$\therefore \text{fog}(x) = f\{g(x)\} = f[|2x + 5|]$$

$$\Rightarrow -5 \leq |2x + 5| \leq 7 \Rightarrow 0 \leq |2x + 5| \leq 7$$

$$\Rightarrow -7 \leq 2x + 5 \leq 7$$

$$\Rightarrow -12 \leq 2x \leq 2$$

$$\Rightarrow -6 \leq x \leq 1$$

$$\therefore \text{fog}(x) \in [-6, 1]$$

Q22 (3) - Composite Functions

Given, $f(x) = \cot^{-1}(x) : \mathbb{R}^+ \rightarrow \left(0, \frac{\pi}{2}\right)$

and $g(x) = 2x - x^2 : \mathbb{R} \rightarrow \mathbb{R}$

$$\therefore f\{g(x)\} = \cot^{-1} g(x)$$

$$= \cot^{-1}(2x - x^2)$$

So, the range of $f\{g(x)\}$ is $\left[\frac{\pi}{4}, \frac{\pi}{2}\right)$

Q23 (1) - Mappings

We have, gof is injective and surjective such that $\text{gof} : A \in C$

Let $h(x) = \text{gof}(x)$

$$\therefore h(y) = \text{gof}(y)$$

$$h(x) = h(y) \Rightarrow \text{gof}(x) = \text{gof}(y)$$

$$g\{f(x)\} = g\{f(y)\} \Rightarrow f(x) = f(y) \quad (\because g \text{ is injective})$$

$$\Rightarrow x = y \quad (\because f \text{ is injective})$$

Thus, $\text{gof}(x) = \text{gof}(y), \forall x, y \in R$

$\therefore \text{gof}$ is injective iff g and f both injective.

Let $z \in C$ such that $g(y) = z$ ($\because g$ is surjective) and

$y \in B$ such that $f(x) = y$ ($\because f$ is surjective)

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$$\therefore g \circ f(x) = g(y) = z$$

Every element of C is the same element of A .

So, $g \circ f$ is surjective iff g and f both are surjective.

Q24 (4) - Mappings

We have, $f(z) = |z|$

$$\text{Let } z_1 = 1 + i, |z_1| = \sqrt{2}$$

$$\text{and } z_2 = 1 - i \quad |z_2| = \sqrt{2}$$

$$\Rightarrow f(z_1) = f(z_2), \quad \text{but } z_1 \neq z_2$$

$\therefore f(z)$ is not one-one function.

Also, $|z| \geq 0$

$\therefore f(z)$ is also not onto function.

Hence, $f(z)$ is neither one-one nor onto function.

Q25 (3) - Mappings

We have, $f(x) = \frac{x^2-4}{x^2+4}$

$$f(x_1) = \frac{x_1^2-4}{x_1^2+4} \text{ and } f(x_2) = \frac{x_2^2-4}{x_2^2+4}$$

$$f(x_1) = f(x_2) \Rightarrow \frac{x_1^2-4}{x_1^2+4} = \frac{x_2^2-4}{x_2^2+4}$$

$$\Rightarrow x_1^2 x_2^2 + 4x_1^2 - 4x_2^2 - 16 = x_1^2 x_2^2 - 4x_1^2 + 4x_2^2 - 16$$

$$\Rightarrow 8x_1^2 - 8x_2^2 = 0$$

$$\Rightarrow x_1^2 - x_2^2 = 0 \Rightarrow (x_1 + x_2)(x_1 - x_2) = 0$$

$$x_1 = -x_2 \text{ or } x_1 = x_2$$

$\therefore f(x)$ is many-one function. Let $y \in R$ such that $f(x) = y$

$$\Rightarrow y = \frac{x^2-4}{x^2+4} \Rightarrow x^2 y + 4y = x^2 - 4$$

$$\Rightarrow x^2(1-y) = 4(y+1)$$

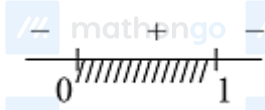
$$\Rightarrow x^2 = \frac{4(y+1)}{1-y}, x^2 \geq 4 \Rightarrow \frac{4(y+1)}{1-y} \geq 4$$

$$\Rightarrow \frac{y+1}{1-y} \geq 1 \quad \frac{y+1-1+y}{1-y} \geq 0 \Rightarrow \frac{y+1}{1-y} - 1 \geq 0$$

$$\Rightarrow \frac{2y}{1-y} \geq 0$$

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$$0 \leq y < 1$$

Hence, the range of $f(x)$ is $[0,1)$ i.e. into

Q26 (3) - Mappings

$$\text{We have, } f(n) = \begin{cases} \frac{n-1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$$

$$\therefore f(3) = \frac{3-1}{2} = 1 \text{ and } f(2) = \frac{2}{2} = 1$$

$$\text{Thus, } f(3) = f(2), \text{ but } 3 \neq 2$$

So, f is many-one function.

Surjectively : Let n be any arbitrary element of N . If n is odd natural number, then $2m - 1$ is also an odd

natural number such that

$$f(2m - 1) = \frac{2m-1-1}{2} = m - 1$$

If n is even natural number, then $2m$ is also an even natural number such that

Thus, for every $m \in N$ (whether even or odd).

So, f is an onto function.

$\therefore f(x)$ is many-one and onto function.

Q27 (4) - Mappings

$$\text{We have, } f(x) = x + \sqrt{x^2}$$

$$\Rightarrow f(x) = x + |x| \left[\because \sqrt{x^2} = |x| \right]$$

$$\Rightarrow f(x) = \begin{cases} 0, & x \leq 0 \\ 2x, & x > 0 \end{cases}$$

$\therefore f(x)$ is neither one-one nor onto function.

Q28 (4) - Mappings

$$\text{We have, } f(n) = (n + 5)^2$$

$$f(n_1) = f(n_2) \Rightarrow (n_1 + 5)^2 = (n_2 + 5)^2$$

$$\Rightarrow n_1 = n_2$$

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$\therefore f(n)$ is one-one function.

Let $y \in N$ is such that $f(n) = y \Rightarrow y = (n + 5)^2$

$\Rightarrow n \notin N$ for $y = 2$. So, f is not onto.

Q29 (1) - Inverse of Function

(1) Given, $f(x) = \frac{1}{x-1}, x \neq 1$

Let $f(x_1) = f(x_2) \Rightarrow \frac{1}{x_1-1} = \frac{1}{x_2-1}$

$\Rightarrow x_1 - 1 = x_2 - 1$

$\Rightarrow x_1 = x_2$

Hence, $f(x)$ is one-one.

Let $y = \frac{1}{x-1} \Rightarrow x - 1 = \frac{1}{y}$

$\Rightarrow x = \frac{1}{y} + 1$

$\therefore \text{Range} \in R - \{0\}$

f is onto, so f^{-1} exists.

(2) Given, $y = x^2 = f(x)$

Let $f(x_1) = f(x_2)$

$\Rightarrow x_1^2 = x_2^2$

$\Rightarrow x_1 = \pm x_2$

f is not one-one, so f^{-1} does not exist.

(3) Given, $f(x) = x^2, x > 0$

$\Rightarrow x_1 = x_2 (x > 0)$

$\therefore f$ is one-one.

Now, $y = x^2$

$x = \sqrt{y} \Rightarrow y > 0$

f is not onto, so f^{-1} does not exist.

(4) Given, $f(x) = x^2, x \leq 0$

Let $f(x_1) = f(x_2)$

$x_1^2 = x_2^2 \Rightarrow x_1 = \pm x_2$

$x_1 = x_2$ (as $x \leq 0$)

$\therefore f$ is one-one.

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$$\text{Range} \Rightarrow y \geq 0$$

f is into, so f^{-1} does not exist.

Q30 (3) - Inverse of Function

$$\text{Given: } f : (2, 4) \rightarrow (1, 3)$$

$$\Rightarrow f(x) = x - \left[\frac{x}{2} \right]$$

$$\Rightarrow f^{-1}(x) = ?$$

$$\Rightarrow f(x) = x - \left[\frac{x}{2} \right]$$

$$\Rightarrow x \in (2, 4)$$

$$\Rightarrow \frac{x}{2} \in (1, 2)$$

$$\Rightarrow [x/2] = 1$$

$$\Rightarrow f(x) = x - 1$$

$$\Rightarrow \text{let } f(x) = y$$

$$\Rightarrow x = f^{-1}(y)$$

$$\Rightarrow y = x - 1$$

$$\Rightarrow x = y + 1$$

$$\Rightarrow f^{-1}(y) = y + 1$$

$$\Rightarrow f^{-1}(x) = x + 1$$

Q31 (2) - Inverse of Function

$$\text{Given } f(x) = \frac{1-x}{1+x^2}$$

$$\text{Let } y = \frac{1-x}{1+x^2} \Rightarrow y + yx^2 = 1 - x$$

$$\Rightarrow yx^2 + x + (y - 1) = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1-4y(y-1)}}{2y} \Rightarrow x = \frac{-1 \pm \sqrt{1+4y(1-y)}}{2y}$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1+4y-4y^2}}{2y}$$

$$\Rightarrow f^{-1}(x) = \frac{-1 \pm \sqrt{1+4x-4x^2}}{2x}$$

$$f^{-1}(x) = \frac{-1 + \sqrt{1+4x-4x^2}}{2x}, x \neq 0$$

and $f^{-1}(x) = 1$, when $x = 0$.

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Q32 (1) - Inverse of Function

Given, $f(x) = \log_a x^2, a > 1$

Let $y = \log_a x^2$, then $a^y = x^2 \Rightarrow x = \sqrt{a^y}$

$\Rightarrow f^{-1}(x) = \sqrt{a^x} \Rightarrow f^{-1}(b+c) = \sqrt{a^{b+c}}$

$f^{-1}(b+c) = \sqrt{a^b \cdot a^c} \dots (i)$

Now, $\left. \begin{aligned} f^{-1}(b) &= \sqrt{a^b} \\ f^{-1}(c) &= \sqrt{a^c} \end{aligned} \right\} \dots (ii)$

Using Eq. (ii) in Eq. (i), we get

$f^{-1}(b+c) = f^{-1}(b) \cdot f^{-1}(C)$